

1. An urn initially contains five black balls and five white balls. The following experiment is repeated indefinitely. A ball is drawn from the urn. If the ball is white, it is put back in the urn, otherwise it is left out. Let X_n be the no. of black balls remaining in the urn after n draws from the urn. Is X_n a Markov process? If so,
- find the appropriate transition probabilities.
 - find the one-step transition probability matrix P for X_n .
 - find the two-step transition probability matrix P^2 by matrix multiplication.
 - what happens to X_n as n approaches infinity? use your guess answer to guess the limit of P^n as $n \rightarrow \infty$

Soln The no. X_n of black balls in the urn completely specifies the prob. of the outcomes of a trial.
 $\therefore$ X_n is independent of its past values and X_n is a Markov process.

$$\begin{aligned}
 \text{a) } P(X_n=4/X_{n-1}=5) &= \frac{5}{10} = 1 - P(X_n=5/X_{n-1}=5) \\
 P(X_n=3/X_{n-1}=4) &= \frac{4}{9} = 1 - P(X_n=4/X_{n-1}=4) \\
 P(X_n=2/X_{n-1}=3) &= \frac{3}{8} = 1 - P(X_n=3/X_{n-1}=3) \\
 P(X_n=1/X_{n-1}=2) &= \frac{2}{7} = 1 - P(X_n=2/X_{n-1}=2) \\
 P(X_n=0/X_{n-1}=1) &= \frac{1}{6} = 1 - P(X_n=1/X_{n-1}=1) \\
 P(X_n=0/X_{n-1}=0) &= 1
 \end{aligned}$$

Transition probability matrix is

b)

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{6} & \frac{5}{6} & 0 & 0 & 0 & 0 \\ 0 & \frac{2}{7} & \frac{5}{7} & 0 & 0 & 0 \\ 0 & 0 & \frac{3}{8} & \frac{5}{8} & 0 & 0 \\ 0 & 0 & 0 & \frac{4}{9} & \frac{5}{9} & 0 \\ 0 & 0 & 0 & 0 & \frac{5}{10} & \frac{5}{10} \end{bmatrix} \end{matrix}$$

c) The two-step tpm, P^2 by matrix multiplication is

$$P^2 = P \cdot P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ \frac{11}{36} & \frac{25}{36} & 0 & 0 & 0 & 0 \\ \frac{1}{21} & \frac{65}{144} & \frac{25}{49} & 0 & 0 & 0 \\ 0 & \frac{3}{28} & \frac{225}{448} & \frac{25}{64} & 0 & 0 \\ 0 & 0 & \frac{1}{6} & \frac{95}{192} & \frac{25}{81} & 0 \\ 0 & 0 & 0 & \frac{2}{9} & \frac{19}{36} & \frac{1}{4} \end{bmatrix} \end{matrix}$$

d) As $n \rightarrow \infty$ eventually all the black balls are removed.

Thus,

$$P^n = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

2. A raining process is considered as a two state Markov chain. If it rains, it is considered to be in state 0, and it doesn't rain, the chain is in the state of 1. The transition probability of the Markov chain is defined by $P = \begin{bmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{bmatrix}$. Find the prob. that it will rain for 3 days from today assuming that it is raining today.

Sol) $P = \begin{bmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{bmatrix}$

$$P(2) = P^2 = \begin{bmatrix} 0.44 & 0.56 \\ 0.28 & 0.72 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} \underline{0.376} & 0.624 \\ 0.312 & 0.688 \end{bmatrix}$$

The prob. that it will rain on third day given that it will rain today is 0.376.

3. suppose that the prob. of a dry day (state 0) follows a rain day (state 1) is $1/3$. and the prob. of a rain day follows dry day is $1/2$. Then find the prob. ~~that~~ ^{Given} If March 1st is a dry day, then find the prob. that [^] i) March 3rd also a dry day, ii) March 5th also a dry day.

Sol From the above problem, we have a two state Markov chain such that $P_{10} = 1/3$, and $P_{01} = 1/2$ and transition prob. matrix (tpm) is given by

$$P = \begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{bmatrix} 1/2 & 1/2 \\ 1/3 & 2/3 \end{bmatrix} \end{matrix} \quad \begin{matrix} \text{rainy to dry} \\ \text{dry to rainy} \end{matrix}$$

1 to 0 is $1/3$

$$P^2 = P \cdot P = \begin{bmatrix} 1/2 & 1/2 \\ 1/3 & 2/3 \end{bmatrix} \cdot \begin{bmatrix} 1/2 & 1/2 \\ 1/3 & 2/3 \end{bmatrix} = \begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{bmatrix} 5/12 & 7/12 \\ 7/18 & 11/18 \end{bmatrix} \end{matrix}$$

and $P^4 = \begin{bmatrix} \underline{173} & 259 \\ 432 & 432 \\ \underline{259} & 389 \\ 432 & 432 \end{bmatrix}$

From this we can conclude that if March 1st is a dry day then the prob. that March 3rd is a dry day is $5/12$ & March 5 is a dry day is $173/432$.

4. A gambler has Rs. 2. He bets Rs. 1 at a time and wins Rs. 1 with prob $\frac{1}{2}$. He stops playing if he loses Rs. 2 or wins Rs. 4.

- what is the tpm of the related Markov chain?
- what is the prob. that he has lost his money at the end of 5 plays?
- what is the prob. that the game lasts more than 7 plays?

Sol Let X_n represents the amount with the player at the end of n th round of the play.

The state space of $X_n = \{0, 1, 2, 3, 4, 5, 6\}$

The game is stopped if the player loses Rs. 2 i.e. $X_n = 0$ or if he wins Rs. 4 i.e. $X_n = 6$.

The tpm of the Markov chain is written as.

	0	1	2	3	4	5	6
0	1	0	0	0	0	0	0
1	$\frac{1}{2}$	0	$\frac{1}{2}$	0	0	0	0
2	0	$\frac{1}{2}$	0	$\frac{1}{2}$	0	0	0
3	0	0	$\frac{1}{2}$	0	$\frac{1}{2}$	0	0
4	0	0	0	$\frac{1}{2}$	0	$\frac{1}{2}$	0
5	0	0	0	0	$\frac{1}{2}$	0	$\frac{1}{2}$
6	0	0	0	0	0	0	1

$0 \rightarrow 0$
 chance is 1
 $\therefore$ If he has no amount he will not play, but has 0 only with chance = 1.
 from $0 \rightarrow 1, 2$ is not possible
 $1 \rightarrow 1$ is not possible
 $6 \rightarrow 6$ because opponent has no money to play, certainly player will have 6 Rs only.

b) Since the player has Rs 2 to start play, the initial prob. distribution of X_n is given by

$$p^{(0)} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$p^{(1)} = p^{(0)} \cdot P = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \dots \end{bmatrix}$$

$$= \begin{bmatrix} 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 \end{bmatrix}$$

$$p^{(1)} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \dots \end{bmatrix}$$

$$\begin{bmatrix} 0 \times 1 + 0 \times \frac{1}{2} + 1 \times 0 + 0 \times 0 + 0 \times 0 + 0 \times 0 + 0 \times 0, & 0 \times 0 + 0 \times 0 + 1 \times \frac{1}{2} + 0 \times 0 + 0 \times 0 + 0 \times 0 + 0 \times 0 \\ 0 \times 0 + 0 \times \frac{1}{2} + 1 \times 0 + 0 \times \frac{1}{2} + 0 \times 0 + 0 \times 0 + 0 \times 0, & 0 \times 0 + 0 \times 0 + 1 \times \frac{1}{2} + 0 \times 0 + 0 \times 0 + 0 \times 0 + 0 \times 0 \\ 0 \times 0 + 0 \times 0 + 1 \times 0 + 0 \times 0 + 0 \times 0 + 0 \times 0 + 0 \times 0, & 0 \times 0 + 0 \times 0 + 1 \times 0 + 0 \times 0 + 0 \times 0 + 0 \times 0 + 0 \times 0 \\ 0 \times 0 + 0 \times 0 + 1 \times 0 + 0 \times 0 + 0 \times 0 + 0 \times 0 + 0 \times 0, & 0 \times 0 + 0 \times 0 + 1 \times 0 + 0 \times 0 + 0 \times 0 + 0 \times 0 + 0 \times 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Hence } p^{(2)} = p^{(1)} \cdot P = \begin{bmatrix} 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \dots \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{4} & 0 & \frac{1}{2} & 0 & \frac{1}{4} & 0 & 0 \end{bmatrix}$$

$$p^{(3)} = p^{(2)} \cdot P = \begin{bmatrix} \frac{1}{4} & 0 & \frac{1}{2} & 0 & \frac{1}{4} & 0 & 0 \end{bmatrix} \begin{bmatrix} \dots \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & 0 & \frac{3}{8} & 0 & \frac{1}{8} & 0 \end{bmatrix}$$

$$p^{(4)} = p^{(3)} \cdot P = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & 0 & \frac{3}{8} & 0 & \frac{1}{8} & 0 \end{bmatrix} \begin{bmatrix} \dots \end{bmatrix}$$

$$P^{(4)} = \begin{bmatrix} \frac{3}{8} & 0 & \frac{5}{16} & 0 & \frac{1}{4} & 0 & \frac{1}{16} \end{bmatrix}$$

$$P^{(5)} = P^{(4)} \cdot P = \begin{bmatrix} \frac{3}{8} & 0 & \frac{5}{16} & 0 & \frac{1}{4} & 0 & \frac{1}{16} \end{bmatrix} \begin{bmatrix} \vdots \end{bmatrix}$$

$$= \begin{bmatrix} \frac{3}{8} & \frac{5}{32} & 0 & \frac{9}{32} & 0 & \frac{1}{8} & \frac{1}{64} \end{bmatrix}$$

The prob. that the player has lost his money at the end of 5 plays = $P(X_5=0) = \frac{3}{8}$.

c) The prob that the game lasts more than 7 plays
 = $P\{\text{system is neither in state 0 nor in 6 at the end of seventh round}\}$.

$$\text{viz } P^{(6)} = P^{(5)} \cdot P = \begin{bmatrix} \frac{3}{8} & \frac{5}{32} & 0 & \frac{9}{32} & 0 & \frac{1}{8} & \frac{1}{64} \end{bmatrix} \begin{bmatrix} \vdots \end{bmatrix}$$

$$= \begin{bmatrix} \frac{29}{64} & 0 & \frac{7}{32} & 0 & \frac{13}{64} & 0 & \frac{1}{8} \end{bmatrix}$$

$$\& P^{(7)} = P^{(6)} \cdot P = \begin{bmatrix} \frac{29}{64} & \frac{7}{64} & 0 & \frac{27}{128} & 0 & \frac{13}{128} & \frac{1}{8} \end{bmatrix}$$

$$\therefore P\{X_7 = 1, 2, 3, 4 \text{ or } 5\} = \frac{7}{64} + 0 + \frac{27}{128} + 0 + \frac{13}{128}$$

$$= \frac{27}{64}$$

Gambler Ruin problem:-

Suppose a Gambler wins or loses a unit (Rupee or dollar or any) with prob. p & q respectively. Let his initial capital be Z and his opponent's initial capital be $(a-Z)$ so that the combined capital be a . The game continues until the gambler's capital is either reduced to zero or increased to a , i.e. until one of the 2 players is ruined.

Here we are interested in finding prob. of the gambler ruin and the prob. distribution of the duration of the game.

Here the capital of the gambler after ' n ' stages is a random variable.

Let q_z denote the prob. of ultimate ruin of the gambler. After the first trial, the gambler's capital is $(Z+1)$ if he wins (with prob. p) or it is $(Z-1)$ if he loses (with prob. q).

The prob. of his ruin after one trial will be q_{Z+1} and q_{Z-1} respectively.

~~Case 1~~ $p \neq q$

The prob. p_z of opponent's ruin can be obtained as

~~Eq~~ The ~~prob~~

$$\text{The prob of ruin } q_z = \frac{\left(\frac{q}{p}\right)^a - \left(\frac{q}{p}\right)^z}{\left(\frac{q}{p}\right)^a - 1}$$

Expected duration of the game

$$d_z = -\frac{a}{q-p} \left[\frac{\left(\frac{q}{p}\right)^z - 1}{\left(\frac{q}{p}\right)^a - 1} \right] + \frac{z}{q-p}$$

Case ii) If $p=q=\frac{1}{2}$.

$$\text{prob of ruin } = q_z = \frac{a-z}{a} \text{ or } 1 - \frac{z}{a}$$

$$\text{Expected duration of the game } = d_z = z(a-z)$$

i) Calculate the prob. of ruin and expected duration of game, when.

$$\text{i) } a=100, p=0.6, z=5. \quad \text{ii) } a=50, z=40, p=0.5$$

$$\text{i) } p=0.6, q=0.4 \Rightarrow \frac{q}{p} = \frac{0.4}{0.6} = \frac{2}{3}$$

$$\begin{aligned} \text{prob of ruin } q_z &= \frac{\left(\frac{q}{p}\right)^a - \left(\frac{q}{p}\right)^z}{\left(\frac{q}{p}\right)^a - 1} \\ &= \frac{\left(\frac{2}{3}\right)^{100} - \left(\frac{2}{3}\right)^5}{\left(\frac{2}{3}\right)^{100} - 1} \approx 0.132 \end{aligned}$$

Expected duration of the game

$$d_z = -\frac{a}{q-p} \left(\frac{\left(\frac{q}{p}\right)^z - 1}{\left(\frac{q}{p}\right)^a - 1} \right) + \frac{z}{q-p}$$

$$dz = \frac{-100}{(0.4-0.6)} \left[\frac{\left(\frac{2}{3}\right)^5 - 1}{\left(\frac{2}{3}\right)^{100} - 1} \right] + \frac{5}{(0.4-0.6)}$$

$$= 500 \left[\frac{\left(\frac{2}{3}\right)^5 - 1}{\left(\frac{2}{3}\right)^{100} - 1} \right] - 25$$

$$= 409$$

(iii) $p=q=\frac{1}{2}$, $a=50$, $z=40$.

$$q_z = 1 - \frac{z}{a} = 1 - \frac{40}{50} = \frac{1}{5} = 0.2.$$

$$dz = z(a-z) = 40(50-40) = 400.$$

Q. The transition probability matrix of a Markov chain is given by

$$\begin{bmatrix} 0.3 & 0.7 & 0 \\ 0.1 & 0.4 & 0.5 \\ 0 & 0.2 & 0.8 \end{bmatrix}.$$

Is this matrix irreducible?

Sol:

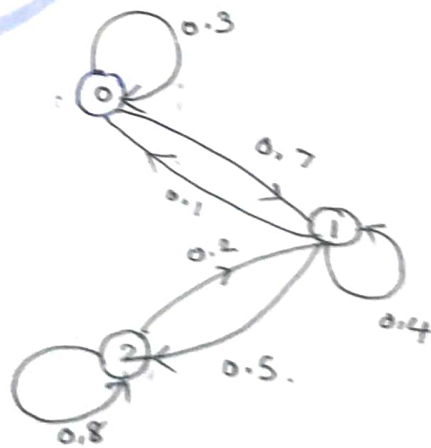
$$\begin{matrix} & \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \end{matrix} & \begin{bmatrix} 0.3 & 0.7 & 0 \\ 0.1 & 0.4 & 0.5 \\ 0 & 0.2 & 0.8 \end{bmatrix} \end{matrix}$$

considers 3 states 0, 1, 2.

In this chain we go from state 0 to state 1 with prob of 0.7 and from state 1 to state 2 with prob 0.5.

Thus it is possible to go from state 0 to state 2.

$\therefore$ The chain is irreducible and all the states are recurrent.



3. If the transition prob. matrix shares of 3 bonds A, B and C is
$$P_{ij} = \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.8 & 0.1 & 0.1 \\ 0.35 & 0.25 & 0.4 \end{bmatrix}$$
 and the initial market

shares are 50%, 25% & 25%.

find (a) The market shares in 2nd & 3rd periods.

(b) limiting probabilities.

Soln

$$P_{ij} = \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.8 & 0.1 & 0.1 \\ 0.35 & 0.25 & 0.4 \end{bmatrix}$$

$$P^{(0)} = \begin{bmatrix} 0.5 & 0.25 & 0.25 \end{bmatrix}$$

$$P^{(1)} = P^{(0)} \cdot P = \begin{bmatrix} 0.5 & 0.25 & 0.25 \end{bmatrix} \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.8 & 0.1 & 0.1 \\ 0.35 & 0.25 & 0.4 \end{bmatrix}$$

$$= \begin{bmatrix} 0.4875 & 0.2375 & 0.275 \end{bmatrix}$$

$$P^{(2)} = P^{(1)} \cdot P = \begin{bmatrix} 0.4875 & 0.2375 & 0.275 \end{bmatrix} \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.8 & 0.1 & 0.1 \\ 0.35 & 0.25 & 0.4 \end{bmatrix}$$

$$= \begin{bmatrix} 0.4812 & 0.2387 & 0.28 \end{bmatrix}$$

$$P^{(3)} = P^{(2)} \cdot P = \begin{bmatrix} 0.4812 & 0.2387 & 0.28 \end{bmatrix} \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.8 & 0.1 & 0.1 \\ 0.35 & 0.25 & 0.4 \end{bmatrix}$$

$$= \begin{bmatrix} \quad \quad \quad \end{bmatrix} \quad \text{Find out}$$

(b) The ~~limiting~~ limiting probabilities.

- (b) The homogeneous Markov chain will have a tpm that is independent of initial state i and steps n as $n \rightarrow \infty$ and is called steady state probability i.e. $q_j = \lim_{n \rightarrow \infty} P_{ij}^{(n)}$ and $\sum_j q_j = 1$ where q_j is called limiting state probability and is interpreted as the long run proportion of time the Markov chain spends in state j .

$$q_j = \lim_{n \rightarrow \infty} P_{ij}^{(n)} \quad \& \quad \sum_j q_j = 1.$$

$$P_{ij}^{(2)} = \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.8 & 0.1 & 0.1 \\ 0.35 & 0.25 & 0.4 \end{bmatrix} \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.8 & 0.1 & 0.1 \\ 0.35 & 0.25 & 0.4 \end{bmatrix} = \begin{bmatrix} 0.505 & 0.225 & 0.27 \\ 0.435 & 0.275 & 0.29 \\ 0.48 & 0.23 & 0.29 \end{bmatrix}$$

$$P_{ij}^{(3)} = \begin{bmatrix} 0.4765 & 0.2415 & 0.282 \\ 0.4955 & 0.2303 & 0.274 \\ 0.4775 & 0.2395 & 0.283 \end{bmatrix}$$

$$P_{ij}^{(4)} = \begin{bmatrix} & & \end{bmatrix}, \quad P_{ij}^{(5)} = \begin{bmatrix} & & \end{bmatrix}, \quad P_{ij} = \begin{bmatrix} & & \end{bmatrix}$$

$$P_{ij}^{(7)} = \begin{bmatrix} 0.4813 & 0.2383 & 0.2803 \\ 0.4813 & 0.2383 & 0.2803 \\ 0.4813 & 0.2383 & 0.2803 \end{bmatrix} = P_{ij}^{(8)} = \dots = P_{ij}^{(n)} \text{ as } n \rightarrow \infty$$

So, we stop the process and take

limiting state probabilities as

$$q_j = 0.4813 \quad 0.2383 \quad 0.2803. \quad \& \quad \sum q_j = 1.$$

4. A fair die is tossed repeatedly. If X_n denotes the maximum of the numbers occurring in the first n tosses, find the transition prob. matrix P of the Markov chain $\{X_n\}$. Find also P^2 & $P[X_2=6]$

Sol Here statespace is $\{1, 2, 3, 4, 5, 6\}$.

Let X_n represents max. of the numbers occurring in the first 'n' trials = 3 (say).

Then $X_{n+1} = 3$ if $(n+1)^{th}$ trial results in 1, 2 or 3.

" $= 4$ if " " " " 4
 " $= 5$ " " " " 5
 " $= 6$ " " " " 6.

$$\therefore P(X_{n+1}=3 / X_n=3) = \overset{P(1)}{\frac{1}{6}} + \overset{P(2)}{\frac{1}{6}} + \overset{P(3)}{\frac{1}{6}}.$$

$$P(X_{n+1}=i / X_n=3) = \frac{1}{6} \text{ when } i=4, 5, 6.$$

$\therefore$ Transition prob matrix is

$$P = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} & \begin{matrix} 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 \\ 0 & 2/6 & 1/6 & 1/6 & 1/6 & 1/6 \\ 0 & 0 & 3/6 & 1/6 & 1/6 & 1/6 \\ 0 & 0 & 0 & 4/6 & 1/6 & 1/6 \\ 0 & 0 & 0 & 0 & 5/6 & 1/6 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{matrix} \end{bmatrix}$$

$$P^2 = \frac{1}{36} \begin{bmatrix} 1 & 3 & 5 & 7 & 9 & 11 \\ 0 & 4 & 5 & 7 & 9 & 11 \\ 0 & 0 & 9 & 7 & 9 & 11 \\ 0 & 0 & 0 & 16 & 9 & 11 \\ 0 & 0 & 0 & 0 & 25 & 11 \\ 0 & 0 & 0 & 0 & 0 & 36 \end{bmatrix}$$

Max is 2 so
 getting 1 is
 impossible
 so 0

$$\begin{aligned} P\{X_{n+1}=2 / X_n=2\} &= \overset{P(1)}{\frac{1}{6}} + \overset{P(2)}{\frac{1}{6}} \\ &= \frac{2}{6} \end{aligned}$$

Initial state prob. distribution is

$$p^{(0)} = \left(\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6} \right) \quad \because \text{all the values are equally likely.}$$

$$\therefore P(X_2=6) = \sum_{i=1}^6 P\{X_2=6/X_0=i\} \times P\{X_0=i\}$$

Total probability

$$= \sum_{i=1}^6 P_{i6}^{(2)} \times \frac{1}{6}$$

$$= \frac{1}{6} \times \frac{1}{36} (11+11+11+11+11+11)$$

$$= \frac{91}{216}$$

$$P(A) = \sum P(E_i) P(A|E_i)$$

5. The transition prob. matrix of a Markov chain $\{X_n\}$ $n=1,2,3,\dots$ having 3 states 1, 2 & 3 is

$$P = \begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{bmatrix} \text{ and the initial distribution}$$

$$\text{is } p^{(0)} = (0.7, 0.2, 0.1).$$

$$\text{find (i) } P\{X_2=3\} \quad \text{ii) } P\{X_3=2, X_2=3, X_1=3, X_0=2\}.$$

sol we have $P(X_0=1)=0.7$, $P(X_0=2)=0.2$

$$P(X_0=3)=0.1.$$

$$p^{(2)} = P^2 = \begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{bmatrix} \begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{bmatrix} = \begin{matrix} 1 & 2 & 3 \\ \begin{bmatrix} 0.43 & 0.31 & 0.26 \\ 0.24 & 0.42 & 0.34 \\ 0.36 & 0.35 & 0.29 \end{bmatrix} \end{matrix}$$

$$P\{X_2=3\} = \sum P\{X_2=3/X_0=i\} \times P\{X_0=i\}$$

$$= \sum_{i=1}^3 P_{i3}^{(2)} \times P\{X_0=i\}$$

$$= P_{13}^{(2)} P\{X_0=1\} + P_{23}^{(2)} P\{X_0=2\} + P_{33}^{(2)} P\{X_0=3\}$$

$$= 0.26 \times 0.7 + 0.34 \times 0.2 + 0.29 \times 0.1 = 0.279$$

Here $X_0=1$
 $X_2=3$
 so, 2 steps
 P^2

$$(ii) P\{X_1=3/X_0=2\} = P_{23} = 0.2$$

$$P\{X_1=3, X_0=2\} = P\{X_1=3/X_0=2\} \times P\{X_0=2\}$$

$$= 0.2 \times 0.2 = 0.04$$

①

$$\text{Now } P\{X_2=3, X_1=3, X_0=2\} =$$

$$P\{X_2=3/X_1=3, X_0=2\} \times P\{X_1=3, X_0=2\}$$

$$= P\{X_2=3/X_1=3\} \times P\{X_1=3, X_0=2\}$$

by Markov property

$$= P_{33} \times 0.04 \quad \text{for } ①$$

$$= 0.3 \times 0.04 = 0.012 \quad \text{--- } ②$$

$$\text{Now } P\{X_3=2, X_2=3, X_1=3, X_0=2\} =$$

$$= P\{X_3=2/X_2=3, X_1=3, X_0=2\} \times$$

$$P\{X_2=3, X_1=3, X_0=2\}$$

$$= P\{X_3=2/X_2=3\} \times P\{X_2=3, X_1=3, X_0=2\}$$

by Markov property

$$= \cancel{0.4} \cdot \cancel{0.2} \cdot P_{32} \times 0.012 \quad \text{for } ②$$

$$= 0.4 \times 0.012$$

$$= 0.0048$$

Here $X_1=3$
 $X_2=3$

do one step
we are using only

P.